SignMuon, MuonSign, and the Role of Error Feedback

Anonymous submission

Abstract

SignMuon compresses the Muon update to one bit per parameter by taking its elementwise sign: the most direct way to run a matrix-aware optimizer on an extremely low communication budget. It outperforms SignSGD in practice, and yet it can ascend on a linear function. Signing the gradient *before* the Linear Minimization Oracle (LMO) rather than after does not repair this: we exhibit small explicit instances on which sign-before (MuonUSign) and sign-on-both-sides (MuonSign) ascend as well, at every step size and every momentum, so that no placement of the sign around the oracle descends in general. Error feedback, the standard repair for a biased compressor, does not rescue sign-after either: applied to the oracle's output it fails for every smoothness constant, step size and momentum. Applied to the gradient it does work, and EF21-MuonUSign and EF21-MuonSign attain the standard $\mathcal{O}(T^{-1/2})$ rate for the squared gradient norm on smooth nonconvex problems, the latter at one bit in each direction. Experiment then reverses the ordering: across centralized CIFAR-10, federated CIFAR-10 and the nanoGPT speedrun the strongest compressed method is consistently sign-*after*-the-LMO, precisely the placement we prove unrepairable, with the provably convergent variants trailing it. Keeping Muon's update intact, a heuristic, is worth more at these scales than the guarantee is. We document this tension rather than resolve it: the counterexamples constrain worst-case problems, not CIFAR-10.

1 Introduction

Training a deep network across clients consumes bandwidth as well as computation: every round each client transmits a full update, and under a limited link those updates dominate the wall-clock cost. Compression is the standard remedy (Bernstein et al. 2018, 2019; Beznosikov et al. 2023), and SignSGD is its extreme point: one bit per coordinate, at slight cost in accuracy. SignSGD, however, flattens each weight matrix into a vector, discarding structure that other optimizers exploit. Muon exploits precisely that structure, orthonormalizing the momentum matrix before stepping, and in several settings surpasses tuned adaptive methods (Jordan et al. 2024b; Bernstein and Newhouse 2024; Shah et al. 2025). What it transmits, though, is a *dense* matrix at full precision: thirty-two bits per parameter where the budget allows one. Matrix geometry and a one-bit budget are therefore hard to have at once.

We study the natural ways to combine sign compression with the Muon LMO at one bit per parameter: **SignMuon**, which signs *after* the LMO, $\mathrm{sign}(\mathrm{LMO}(\cdot))$; **MuonUSign**, which signs *before*, $\mathrm{LMO}(\mathrm{sign}(\cdot))$; and **MuonSign**, which signs on *both* sides and, like SignMuon, emits a ± 1-valued step, so that a sign needs no second sign and uplink and downlink alike cost one bit. All three build Muon's matrix-aware geometry into the step without preserving it intact, and they are not interchangeable: on federated CIFAR-10 (Table 2) they span several accuracy points, in the order after, before, both sides, with only sign-after matching full-precision Muon and both sides falling below SignSGD.

None of them, however, descends in general. We prove that each can *ascend* on a linear objective, the simplest smooth function there is, at every step size and every momentum: SignMuon on a 4×4 gradient (Theorem 1), MuonUSign and MuonSign on a single 5×5 one (Theorems 2–3), both shapes minimal. The standard repair for a biased compressor is error feedback, and for sign-after it is unavailable: applied to the oracle's *output* it leaves the method divergent for every triple (L, η, μ) (Theorem 4), so no step-size rule built from the smoothness and momentum constants can save it.

What error feedback does repair is the placement that compresses the gradient instead. Adapting EF21-Muon (Gruntkowska et al. 2025) to sign compression gives **EF21-MuonUSign**, which drives $\min_{t \le T} \mathbb{E}\|\nabla f(\mathbf{X}_t)\|_*^2$ to zero at the standard $\mathcal{O}(T^{-1/2})$ rate for smooth nonconvex problems, at a one-bit uplink, and **EF21-MuonSign**, which adds a second error-feedback loop on the downlink so that both channels carry one bit for little further cost. Both descend on the counterexamples above. To our knowledge the sign-before and both-sides placements are new, as are the two error-feedback methods; SignMuon itself was proposed concurrently by Mishra, Trivedi, and Kumar (2026), whose guarantee does not cover the instances of Theorem 1 (Appendix A.7).

We evaluate all six against Muon, SignSGD, SGD and Adam on centralized CIFAR-10 with a ResNet-18, federated CIFAR-10 at $N = 11$ clients, the nanoGPT speedrun, and a synthetic convex quadratic on which the quantity the counterexamples attack can be measured directly. Theory and practice then part company. On random quadratics the gradient/step alignment that Theorems 1–3 drive negative stays positive for all three placements, so the constructions describe a worst case that ordinary data does not produce; and on all three networks the best compressed method is sign-after, exactly the placement we prove unrepairable, the two provably convergent variants trailing it by several seed spreads. Preserving Muon's update heuristically is, at these scales, worth more than the guarantee. A tuned five-seed federated comparison establishes that ordering and locates the cost of the guarantee at the placement rather than at the downlink. Proofs, algorithms and every numbered result cited but not stated here are in the supplementary material, which continues this numbering.

2 Related Work

Sign compression. The sign is the most used of the compressors in this literature (Alistarh et al. 2017; Horváth et al. 2023; Beznosikov et al. 2023): one bit per coordinate, no index set or scale sent beside it, and a majority vote of signs that is again a sign, so both directions stay at one bit (Bernstein et al. 2019). Its theory is correspondingly well developed. SignSGD (Bernstein et al. 2018) needs growing batches to converge; its bias is otherwise repaired by error feedback (Seide et al. 2014; Karimireddy et al. 2019), sharpened into EF21 (Richtárik, Sokolov, and Fatkhullin 2021), by momentum (Cutkosky and Mehta 2020; Sun et al. 2023), or by randomizing the sign operator (Chen et al. 2020; Safaryan and Richtarik 2021; Jin et al. 2025).

LMO optimizers. Muon (Jordan et al. 2024b) is the spectral-norm instance of a norm-constrained LMO step (Bernstein and Newhouse 2024; Pethick et al. 2025), analysed by Li and Hong (2025) and Kovalev (2025), then generalized to arbitrary layer norms by Gluon (Riabinin et al. 2025).

Compressed and federated Muon. Gruntkowska et al. (2025) give the error-feedback framework for bidirectionally compressed Muon/Gluon that EF21-MuonUSign and EF21-MuonSign instantiate. Around it: Qian et al. (2026) compress Gluon with SARAH-type variance reduction; Takezawa et al. (2026) and Zhang and Gao (2025) study federated LMO steps without compression; Ahn et al. (2025) make the orthonormalization low-rank with error feedback, Gupta et al. (2025) quantize Muon's optimizer states, Thérien et al. (2025) quantize the delta of DiLoCo's Muon inner loop (Douillard et al. 2024) to two bits at no loss. Concurrently, Mishra, Trivedi, and Kumar (2026) proposed plain SignMuon with an $\mathcal{O}(1/\sqrt{T})$ guarantee, which Appendix A.7 shows to be no guarantee for their method: it is proved only for a gradient-sign update, while for the polar-sign one an unbounded residual term leaves the inequality vacuous exactly where SignMuon ascends. Two further methods relate sign descent to Muon differently. Bolatov et al. (2026) *alternate* spectral and sign steps rather than composing them; Kravatskiy et al. (2025) mix the geometries *inside* the oracle, so S-Muon stays an LMO for an explicit norm, with the attendant theory. Our placements put the sign *around* the oracle, where it is a compressor and not a geometry, so the counterexamples below bear on neither (Appendix A.8).

3 Problem Statement

We consider the stochastic optimization problem

$$\min_{\mathbf{X}\in\mathcal{X}} \{f(\mathbf{X}) := \mathbb{E}_{\xi\sim\mathcal{D}}[f(\mathbf{X};\xi)]\}, \tag{1}$$

where $\mathcal{X}$ is the parameter space (e.g., $\mathbb{R}^d$ or $\mathbb{R}^{m\times n}$) and $f(\cdot\,;\xi) : \mathcal{X} \to \mathbb{R}$ is continuously differentiable and possibly non-convex: $f(\mathbf{X};\xi)$ is the loss of a model $\mathbf{X}$ at a data point $\xi \sim \mathcal{D}$. In the federated setting the data are split across clients, and the training problem becomes

$$\min_{\mathbf{X}\in\mathcal{X}} \Big\{ f(\mathbf{X}) := \tfrac{1}{N}\textstyle\sum_{j=1}^{N} f_j(\mathbf{X}) \Big\}, \tag{2}$$

where $N \geq 1$ is the number of clients and $f_j(\mathbf{X}) = \mathbb{E}_{\xi_j\sim\mathcal{D}_j}[f_j(\mathbf{X};\xi_j)]$ is the loss on the data $\mathcal{D}_j$ held by client $j \in [N] := \{1,\dots,N\}$.

The model is a tuple of layers $\mathbf{X} = [\mathbf{X}_1,\dots,\mathbf{X}_p]$, $\mathbf{X}_i \in \mathbb{R}^{m_i\times n_i}$ with $m_i, n_i \geq 1$, with ∇_i the gradient block of layer i; iterates carry a time index, $\mathbf{X}_t$, and $\mathbf{X}_{t,i}$ is layer i of iterate t. This is Gluon's parameter space (Riabinin et al. 2025) under a single geometry: each layer carries the spectral norm $\|\cdot\|_{2\to2}$, whose dual is the nuclear norm $\|\cdot\|_*$, and we define $\|\nabla f(\mathbf{X})\|_*^2 := \sum_i \|\nabla_i f(\mathbf{X})\|_*^2$; a single matrix parameter is the case $p = 1$. A vector parameter is a block of width one, on which the spectral norm is Euclidean and the oracle only normalizes, so all three methods of Section 4 coincide with SignSGD there and the divergence results below need $\min(m_i, n_i) \geq 2$ (Appendix A.2).

Two assumptions run through the paper.

Assumption 1 (Lower boundedness) *$f(\mathbf{X}) \geq f^*$ for all $\mathbf{X}$; where explicitly invoked, each local $f_j \geq f_j^*$ as well.*

Assumption 2 (Layer-wise smoothness) *For every layer i and all $\mathbf{X}, \mathbf{Y}$,*

$$\|\nabla_i g(\mathbf{X}) - \nabla_i g(\mathbf{Y})\|_* \leq L_i^g \,\|\mathbf{X}_i - \mathbf{Y}_i\|_{2\to2} \tag{3}$$

for $g = f$ (constant L_i) and $g = f_j$ (constant $L_{i,j}$). A weaker layer-wise (L^0, L^1) form, stated in Appendix A.1, is used only in Corollary 2 (Appendix A.10).

A Linear Minimization Oracle (LMO) minimizes a linear form over the unit ball of a norm, returning the steepest-descent direction in that norm (Appendix A.1). Muon takes that norm to be the RMS→RMS operator norm of each matrix layer, $\sqrt{n/m}\,\|\cdot\|_{2\to2}$ on $\mathbb{R}^{m\times n}$, so the oracle is an orthogonalization scaled by $\sqrt{m/n}$: writing $\mathbf{M} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$ for the rank-r singular value decomposition of $\mathbf{M} \in \mathbb{R}^{m\times n}$, the LMO direction is

$$A(\mathbf{M}) = -\mathbf{U}\mathbf{V}^\top \in \mathbb{R}^{m\times n}, \qquad \operatorname{polar}(\mathbf{M}) := -A(\mathbf{M}). \tag{4}$$

We carry that scale in the per-layer step size (7), where the reference implementation also puts it, and the spectral norm in the assumptions. In practice the oracle is applied to a momentum estimate rather than to a gradient; (5) below gives the full iteration. What Muon must transmit is that dense direction, precisely what a bandwidth-limited link cannot accommodate; our object is therefore its matrix-aware geometry at a one-bit budget.

4 Theory

From SignSGD to Sign A. SignSGD was introduced as a compressed SGD: take a method that works, and transmit only the sign of its update. Nothing in that recipe is specific to SGD, so one may try it on any optimizer that emits a direction and hope the compressed method inherits what the direction contributed. For LMO-based optimizers there is reason to hope, their theory being uniform in the geometry: one analysis covers every norm, which enters only through its oracle and a pair of norm-equivalence constants (Pethick et al. 2025; Kovalev 2025; Riabinin et al. 2025). Were signing a direction harmless, it would be harmless across that family, and the member one seeks to compress is Muon.

A second reason to expect it to work is the one that misleads. The sign step is itself an LMO, steepest descent in ℓ_∞ (Bernstein and Newhouse 2024), so signing a Muon direction composes two oracles, each sound alone. The composition is an LMO for *no* norm. An oracle for a norm $\|\cdot\|$ returns $\mathbf{D} = -A(\mathbf{G})$ with $\langle \mathbf{G}, \mathbf{D}\rangle = \max_{\|\mathbf{Y}\|\le 1}\langle \mathbf{G}, \mathbf{Y}\rangle = \|\mathbf{G}\|_{\mathrm{dual}}$, nonnegative since $\mathbf{0}$ is feasible; Theorem 1 exhibits a gradient on which the composition drives that inner product strictly negative. The guarantee is forfeited in the composition, not in either factor.

The Sign A family. Let A be any optimizer built on LMO directions. A Sign A method initializes $\mathbf{M}_0 = \mathbf{0}$ and at each iteration $t \ge 1$ performs

$$\begin{aligned}
&\mathbf{M}_t = \mu \mathbf{M}_{t-1} + (1-\mu)\,\mathbf{G}_t,\\
&\tilde{\mathbf{M}}_t = \mathbf{M}_t \ \text{ or } \ (1-\mu)\mathbf{G}_t + \mu \mathbf{M}_t \ \ \text{(Nesterov)},\\
&\mathbf{D}_t = -A(\tilde{\mathbf{M}}_t), \qquad \mathbf{s}_t = \operatorname{sign}(\mathbf{D}_t) \in \{\pm 1\}^{m\times n},\\
&\mathbf{X}_t = \mathbf{X}_{t-1} - \eta_t \mathbf{s}_t,
\end{aligned} \tag{5}$$

where $\mathbf{X}_t \in \mathbb{R}^{m\times n}$ is the parameter matrix, $\mathbf{G}_t = \nabla f(\mathbf{X}_{t-1}; \xi_t)$ the stochastic gradient at the current iterate, $\mu \in [0,1)$ the momentum coefficient and η_t the learning rate; $A(\cdot)$ is the LMO, and $\operatorname{sign}(\cdot)$ acts element-wise on the structured direction $\mathbf{D}_t$, with $\operatorname{sign}(0)$ resolved to an independent random ± 1 so that the transmitted alphabet stays binary. Momentum is in exponential-moving-average form throughout; the heavy-ball form differs by the positive factor $1/(1-\mu)$, which $\operatorname{sign}(\cdot)$ and the LMO both absorb, so no result below distinguishes them.

Three placements of the sign. Of the three, only SignMuon is a Sign A method. All three keep the momentum of (5) and the update $\mathbf{X}_t = \mathbf{X}_{t-1} - \eta_t\,\mathbf{s}_t$, and differ only in where the sign acts on $\tilde{\mathbf{M}}_t$:

$$\mathbf{s}_t = \begin{cases} \operatorname{sign}\big(\operatorname{polar}(\tilde{\mathbf{M}}_t)\big) & \text{(SignMuon: after)},\\ \operatorname{polar}\big(\operatorname{sign}(\tilde{\mathbf{M}}_t)\big) & \text{(MuonUSign: before)},\\ \operatorname{sign}\big(\operatorname{polar}(\operatorname{sign}(\tilde{\mathbf{M}}_t))\big) & \text{(MuonSign: both)}. \end{cases} \tag{6}$$

The three are Algorithms A2, A4 and A5 (Appendix A.18), all transmitting one bit per parameter on the uplink. On the downlink SignMuon and MuonSign distribute a ± 1-valued object and so cost one bit each way, whereas MuonUSign's server-side $\operatorname{polar}(\cdot)$ is dense and goes at full precision.

4.1 Centralized Learning

Centrally, SignMuon is applied to the matrix-valued parameters ($n_{dim} \ge 2$), while vector parameters (biases, BatchNorm) and the final classification layer are trained with AdamW, the design of Jordan et al. (2024b), whose LMO branch is what Li and Hong (2025) and Riabinin et al. (2025) analyse. We approximate $\operatorname{polar}(\cdot)$ by a 5th-order Newton–Schulz iteration rather than a full SVD (Algorithm A1). One further implementation choice has consequences for every experiment below.

Per-layer step sizes: the unit-gain heuristic. The methods of this paper produce step matrices from two families with *exactly* known Frobenius norms, for a layer ℓ of shape $m \times n$ (fan-out × fan-in): $\|\mathbf{U}\mathbf{V}^\top\|_F = \sqrt{r}$ for the LMO-terminated methods ($r = \min(m,n)$), and $\|\mathbf{s}\|_F = \sqrt{mn}$ for the SIGN-terminated ones. These scale differently with layer shape, so no single global η is appropriate for both families, or across layers within one. We fix the shape dependence a priori and tune only a shape-free base rate η_0, giving layer ℓ the step size $\eta_\ell = \eta_0 \lambda_\ell$ with

$$\lambda_\ell = \frac{\sqrt{m}}{\|\mathbf{s}_\ell\|_F} \quad \Longrightarrow \quad \begin{cases} \lambda_\ell^{\text{LMO}} = \sqrt{\max(1, m/n)},\\ \lambda_\ell^{\text{SIGN}} = 1/\sqrt{n}. \end{cases} \tag{7}$$

The criterion is that every layer's update have the same root-mean-square input–output gain, the average-case form of the spectral scaling condition (Yang, Simon, and Bernstein 2023; Large et al. 2024), so that η_0 is the per-step RMS gain. We treat (7) as a heuristic and apply it uniformly for one reason above all: its LMO branch *reproduces* the aspect-ratio factor already in the reference Muon implementation (Jordan et al. 2024b), so a criterion accepted for one family supplies the counterpart for the other. Appendix A.17 gives the derivation and the measurement selecting the fan-in exponent $\frac{1}{2}$ over μP's 1 (Yang et al. 2021).

4.2 Divergence of the Three Sign Placements

The Muon LMO direction does not survive sign compression, before or after the oracle: in each of the three placements the compressed step can become an *ascent* direction on a smooth objective. We refute the descent property on *linear* objectives, the simplest smooth functions:

$$f(\mathbf{X}) = \langle \mathbf{G}, \mathbf{X}\rangle = \operatorname{Tr}(\mathbf{G}^\top \mathbf{X}), \qquad \nabla f(\mathbf{X}) \equiv \mathbf{G}. \tag{8}$$

Gradient descent and full-precision Muon drive $f \to -\infty$ here, so a rule that instead drives $f \to +\infty$ is unambiguously ascending (Remark 1 in Appendix A.9 restores Assumption 1 without moving any trajectory below). On (8) the step is the constant matrix $\mathbf{s}(\mathbf{G})$ whatever the momentum, so momentum affects neither convergence nor divergence (Proposition 1, Appendix A.3) and

$$f(\mathbf{X}_t) - f(\mathbf{X}_{t-1}) = -\eta_t \left\langle \mathbf{G},\ \mathbf{s}(\mathbf{G})\right\rangle \tag{9}$$

for every $\mu \in [0,1)$ and either momentum rule. A single $\mathbf{G}$ with $\langle \mathbf{G}, \mathbf{s}(\mathbf{G})\rangle < 0$ therefore makes f increase at every step, and such a $\mathbf{G}$ exists for each placement at small size.

Theorems 1–3 (summary). *There exists* $\mathbf{G} \in \mathbb{R}^{4\times 4}$ *with* $\langle \mathbf{G}, \mathbf{s}(\mathbf{G})\rangle < 0$ *for SignMuon, and* $\mathbf{G} \in \mathbb{R}^{5\times 5}$ *with*

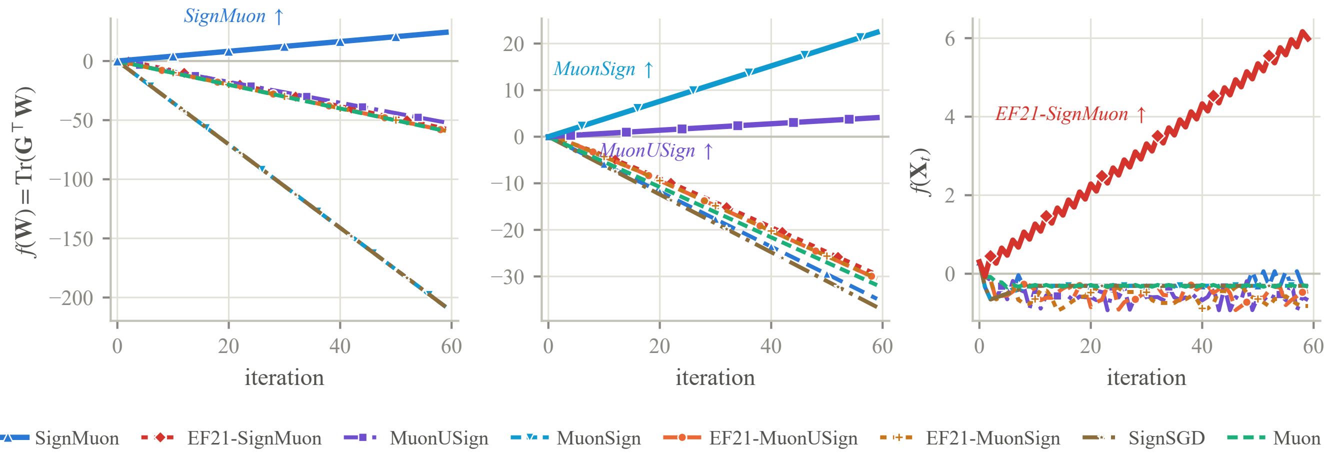

Figure 1: All eight methods on the three counterexample instances; the ascending method is drawn heavy and named in each panel. *Left:* SignMuon, 4×4 (Theorem 1). *Centre:* MuonUSign and MuonSign on their shared 5×5 instance (Theorems 2–3); both panels plot (8). *Right:* EF21-SignMuon, 2×2 (Theorem 4), normalized units at $\eta = 1$. Trajectories are momentum-free without loss (Proposition 1; Lemma 4 for EF21-SignMuon).

$\langle \mathbf{G}, \mathbf{s}(\mathbf{G}) \rangle < 0$ *for MuonUSign and for MuonSign simultaneously. On the corresponding objective* (8) *each method ascends at every iteration, for every* $\eta_t > 0$*, every* $\mu \in [0, 1)$ *and either momentum rule, and* $f(\mathbf{X}_t) \to +\infty$ *whenever* $\sum_t \eta_t = \infty$.

Both instances are explicit and both inner products exact rationals; Appendices A.4–A.6 state the theorems, prove them, and bound the two shapes from below. Figure 1 (left, centre) exhibits the ascents. Negative results of this kind exist for uncompressed Muon as well: Parshakova et al. (2026) show that it need not converge on convex Lipschitz problems. Ours are due to the compressor rather than the geometry, and hold on a smooth objective.

4.3 Centralized Error Feedback: EF21-SignMuon

Theorems 1–3 rule out all three placements of the sign around the oracle. The classical remedy for a biased compressor is *error feedback*: keep what the compressor discarded and fold it into the next message (Seide et al. 2014; Karimireddy et al. 2019), in the EF21 form (Richtárik, Sokolov, and Fatkhullin 2021), which stores a gradient estimator and compresses the *difference* to it, dropping the bounded-gradient assumption. Its most direct use for SignMuon keeps the geometry, the sign *after* the LMO, and applies error feedback to the oracle's output. The resulting method, **EF21-SignMuon** (Algorithm A3), tracks an estimate $\mathbf{d}_t^{\mathrm{est}}$ of the LMO direction $\mathbf{D}_t = \mathrm{polar}(\tilde{\mathbf{M}}_t)$, refreshed by a scaled sign of the residual,

$$\begin{aligned} \mathbf{d}_t^{\mathrm{est}} &= \mathbf{d}_{t-1}^{\mathrm{est}} + \alpha_t \,\mathrm{sign}\big(\mathbf{D}_t - \mathbf{d}_{t-1}^{\mathrm{est}}\big), \\ \alpha_t &= \mathrm{mean}\big|\mathbf{D}_t - \mathbf{d}_{t-1}^{\mathrm{est}}\big|, \end{aligned} \tag{10}$$

and steps $\mathbf{X}_t = \mathbf{X}_{t-1} - \eta_t \,\mathbf{d}_t^{\mathrm{est}}$.

Error feedback does not repair this placement: EF21-SignMuon diverges at *every* step size and momentum setting.

Theorem 4 (Divergence of EF21-SignMuon) *For every* $L > 0$*, step size* $\eta > 0$*, momentum coefficient* $\mu \in [0, 1)$*, and either momentum variant, there is an* L*-smooth (Assumption 2), bounded-below (Assumption 1) function* $f : \mathbb{R}^{2\times 2} \to \mathbb{R}$ *on which EF21-SignMuon started at* $\mathbf{X}_0 = \mathbf{0}$ *diverges: for an explicit constant* $c = c(f) > 0$,

$$f(\mathbf{X}_{t+2}) - f(\mathbf{X}_t) = c\,L\eta^2 > 0 \qquad (t \geq 3), \tag{11}$$

so $f(\mathbf{X}_t) \to +\infty$*. In particular, no step-size rule* $\eta = \eta(L, \mu)$ *using only the smoothness and momentum constants can make the method convergent.*

The construction (Appendix A.9) turns the shared magnitude against the method. Its 2×2 LMO target has a large off-diagonal that reverses sign every step, which keeps the residual — and so α_t — at $\Theta(1)$, and a small constant on the diagonal, which that magnitude overshoots at every step; the diagonal estimate settles into a period-two cycle whose average has the *wrong* sign, and the coordinate it drives runs off to infinity. Dimension is not implicated, 2×2 being where the scaled sign's worst-case contraction $1/d$ is most favourable, nor is momentum: Figure A1 (Appendix A.9) records the same rate for every μ and both variants.

4.4 Centralized Error Feedback: EF21-MuonUSign and EF21-MuonSign

The shared magnitude α_t is not itself the defect: the two methods below couple all coordinates through the same scalar and converge. What fails is the target. Error feedback needs an estimator whose target varies with the step size, and the LMO output $\mathrm{polar}(\tilde{\mathbf{M}}_t)$ is not one: the polar factor is not Lipschitz in its argument, so it can move by $\Theta(1)$ between consecutive rounds however small η is. Compressing the *gradient* in its place restores that dependence, and the mechanism is then EF21-Muon (Gruntkowska et al. 2025), which we instantiate at the sign.

That framework compresses the two directions of the link separately, through a pair $(\mathcal{C}^{\uparrow}, \mathcal{C}^{\downarrow})$ of contractive compressors. Ours in either slot is the scaled sign $\mathcal{C}(\mathbf{Y}) =$

mean $|\mathbf{Y}|$ sign($\mathbf{Y}$): one bit per parameter, plus one scalar per layer. **EF21-MuonUSign** takes the pair $(\mathcal{C}, I)$, a compressed uplink and a full-precision model back, which is the budget to spend when only the uplink is constrained. It maintains a gradient estimator $\mathbf{g}_t^{\mathrm{est}}$, refreshed from the residual $\Delta_t = \tilde{\mathbf{M}}_t - \mathbf{g}_{t-1}^{\mathrm{est}}$:

$$\alpha_t = \mathrm{mean}(|\Delta_t|), \quad \mathbf{g}_t^{\mathrm{est}} = \mathbf{g}_{t-1}^{\mathrm{est}} + \alpha_t \, \mathrm{sign}(\Delta_t), \tag{12}$$

and takes the descent step $\mathbf{X}_t = \mathbf{X}_{t-1} - \eta_t \mathbf{D}_t$ with $\mathbf{D}_t = -A(\mathbf{g}_t^{\mathrm{est}}) \approx \mathbf{U}_t \mathbf{V}_t^\top$. Here $\mathbf{U}_t \mathbf{V}_t^\top$ is the polar factor of the *refreshed* estimator, not of ∇f, so no step-wise descent property is claimed, only the tracking hypothesis the guarantee requires. The sign in (12) acts on the internal compression residual alone, so the uplink stays at one bit per parameter (Algorithm A6).

EF21-MuonSign takes the pair $(\mathcal{C}, \mathcal{C})$: one bit in each direction. Its downlink compressor is a second error-feedback loop, on the *model* rather than the gradient, and it splits the method into two iterates:

$$\begin{aligned} \mathbf{X}_t &= \mathbf{X}_{t-1} - \eta_t \mathbf{D}_t && \text{(server model)}, \\ \mathbf{W}_t &= \mathbf{W}_{t-1} + \alpha_t^{\downarrow} \, \mathrm{sign}(\mathbf{X}_t - \mathbf{W}_{t-1}) && \text{(broadcast model)}, \end{aligned} \tag{13}$$

with $\mathbf{D}_t = -A(\mathbf{g}_t^{\mathrm{est}})$ as before, and $\alpha_t^{\downarrow} = \mathrm{mean}\,|\mathbf{X}_t - \mathbf{W}_{t-1}|$. The server model $\mathbf{X}$ takes the exact LMO step and never leaves the server; what crosses the downlink is the one-bit refresh of $\mathbf{W}$, the only model the rest of the method observes, since gradients, momentum and the uplink residual are all computed there, whereas the guarantee bounds the gradient at $\mathbf{X}$. Under an exact downlink the two coincide and the method reduces to EF21-MuonUSign. The rate survives the second loop, but not without charge: a sign-based downlink contracts in the Euclidean norm, whereas the spectral geometry needs contractivity in the layer norm, so the admissible step size loses a further $\sqrt{r}$ in the layer rank. This is no artefact of the analysis, the scaled sign being spectrally contractive for *no* parameter at all, so only a different compressor removes it (Remark 4, Appendix A.10); Section 5.3 quantifies the expense.

Both methods descend where the placements they repair ascend: Figure 1 (centre) has them on the 5×5 instance of Theorems 2–3. Appendix A.10 shows them to be *exact instances* of the framework, the one substantive check being that the scaled sign is a contractive compressor (Lemma 5), so they inherit its guarantees (Theorem 5). Under Assumptions 1–2 both reach $\min_{t \le T} \mathbb{E}\|\nabla f(\mathbf{X}_t)\|_*^2 = \mathcal{O}(T^{-1/2})$, the standard smooth nonconvex rate, which uncompressed Muon attains as well: one bit costs a constant, not a rate. Under (L^0, L^1)-smoothness EF21-MuonUSign reaches $\min_t \sum_i \mathbb{E}\|\nabla_i f(\mathbf{X}_t)\|_* = \mathcal{O}(T^{-1/4})$ at a constant step size (Corollaries 1–2). Both cost memory as well, one buffer per compressed channel: negligible on the server, possibly an issue on memory-limited workers.

4.5 Federated Learning

In the federated setting (2) the placement decides where the oracle runs. With the sign *after* it, each client must orthogonalize its own momentum: it computes a stochastic gradient at the broadcast model, maintains a momentum buffer, applies the Muon LMO (Algorithm A1), then uploads the elementwise sign of the result. The server takes a majority vote, $\mathbf{s}_t^{\mathrm{agg}} = \mathrm{sign}\big(\sum_j \mathbf{s}_t^{(j)}\big)$, and steps $\mathbf{X}_t = \mathbf{X}_{t-1} - \eta_t \mathbf{s}_t^{\mathrm{agg}}$, as SignSGD does (Bernstein et al. 2019). With the sign *before*, the client uploads $\mathrm{sign}(\tilde{\mathbf{M}}_t^{(j)})$; the server votes, then applies a single LMO to the outcome; the direction that returns is dense, so MuonUSign broadcasts the model at full precision, whereas MuonSign signs that direction once more and broadcasts $\mathrm{sign}(\mathbf{D}_t)$. SignMuon and MuonSign thus send a ± 1-valued object down as well as up — the vote itself in the first case, free of ties at an odd client count — so both directions cost one bit; and since every replica applies the same refresh, copies started from a common $\mathbf{X}_0$ never drift.

Error feedback changes the uplink message, not where the oracle runs: each client sign-compresses the residual between its estimator and the quantity it would otherwise have sent — the LMO output for EF21-SignMuon, the momentum for the other two — and sends $(\mathbf{s}_t^{(j)}, \alpha_t^{(j)})$; the server averages these into a global estimator. The extra per-layer scalar leaves the uplink at $\approx$1 bit per parameter, but that estimator is a scaled average of signs and so is dense, and the vote argument lapses: the full model must be broadcast unless a second error-feedback loop compresses it, as EF21-MuonSign does. The complete procedures are Algorithms A8–A9.

5 Experiments

Does the compressed step descend in practice? On a deterministic convex quadratic (Appendix A.12) we can measure the scalar the descent lemma turns on: the alignment $\rho_t = \langle \nabla f, \mathbf{D}_t \rangle / (\|\nabla f\|_F \|\mathbf{D}_t\|_F)$ between the gradient and the step actually taken. A step descends when $\rho_t > 0$, and Theorems 1–3 force $\rho_t < 0$; on random instances that never happens, all three placements holding ρ_t clear of zero throughout. The counterexamples describe a worst case, not a typical one, which is what lets the network results below run contrary to them. Only EF21-MuonSign dips negative, the one method carrying a proof: error feedback converges through its estimator, not through per-step descent. Two conventions carry into the tables: SignMuon and SignSGD emit steps of *identical* Frobenius length, so what separates them is geometry and not step length; and every method but SGD settles onto an accuracy floor, so a fixed-target iteration count ranks *floors*, not rates.

5.1 Centralized Learning

We compare all six matrix-aware 1-bit methods against Muon, SignSGD, SGD and Adam on CIFAR-10 (Krizhevsky and Hinton 2009) with a ResNet-18 adapted to low-resolution images, over 75 epochs with η_0 cosine-annealed to zero. The only tuned hyperparameter is η_0, selected per method on a held-out split and read identically across methods through the unit-gain rule (7). Centralized numbers average three seeds and federated numbers five; differences below the seed spread are not claimed (Appendix A.11).

The top three rows of Table 1 lie within one seed spread, so we claim only that sign-after *matches* Muon at one bit per parameter, error feedback costing nothing measurable besides.

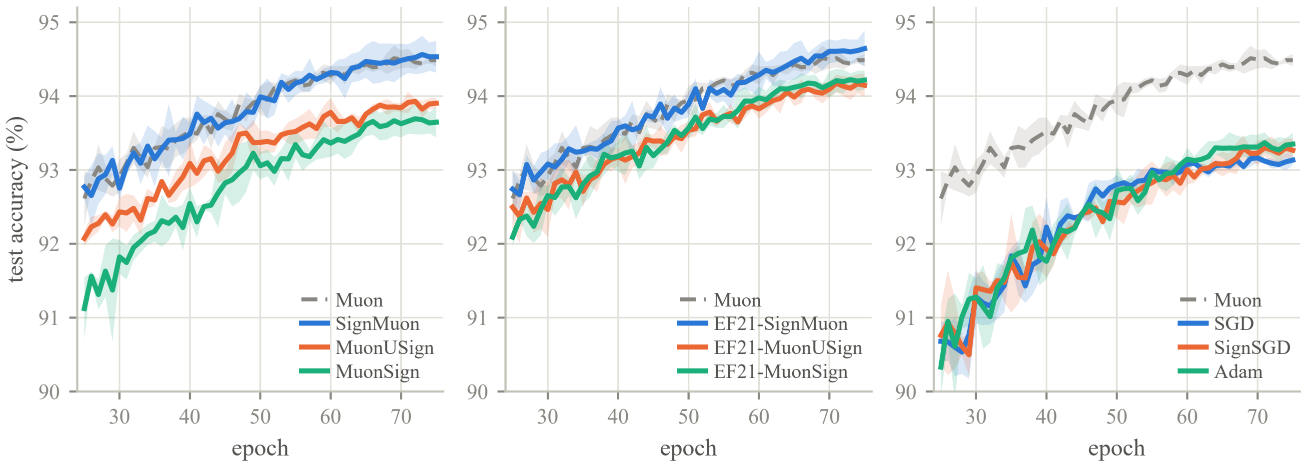

Figure 2: Centralized ResNet-18 on CIFAR-10: test accuracy from epoch 25 at each method's selected η_0 (Table 1). Panels group the sign placements, the EF21 variants and the baselines; Muon is the gray dashed reference, bands ± 1 s.d. over three seeds.

Method	η_0	**Test acc (%)**	**Ep. to 90%**
EF21-SignMuon	0.01	94.64 ± 0.24	8.7
SignMuon	0.02	94.53 ± 0.22	8.3
Muon	0.05	94.49 ± 0.09	7.7
EF21-MuonSign	0.01	94.22 ± 0.13	10.3
EF21-MuonUSign	0.2	94.15 ± 0.16	8.7
MuonUSign	0.02	93.90 ± 0.08	10.0
MuonSign	0.005	93.65 ± 0.21	15.0
Adam	0.0005	93.35 ± 0.15	18.8
SignSGD	0.002	93.27 ± 0.16	19.0
SGD	0.02	93.13 ± 0.07	21.7

Table 1: Centralized ResNet-18 / CIFAR-10, 75 epochs. Test accuracy is the mean $\pm$ s.d. over three seeds of the last five epochs; "Ep. to 90%" is the mean epoch at which it first reaches 90%.

What does resolve is the comparison against SignSGD, which spends the same budget without the LMO: that margin measures the geometry, not the compressor. Within the family the ordering exceeds the noise, after before both sides, and the training loss orders the methods identically (Figure A5). The threshold column separates them far more sharply: SignMuon reaches 90% in 8.3 epochs against 18.8–21.7 for SGD, SignSGD and Adam, and orthogonalization's overhead still leaves a $\sim 2\times$ wall-clock margin.

5.2 Federated Learning

We compare the same ten methods on CNN2 (two convolutional blocks with BatchNorm and an MLP classifier), with CIFAR-10 split homogeneously across $N = 11$ clients and no local steps; N is odd so that the majority vote is decisive in every coordinate. Rates are tuned per method on a split held out *before* the client partition; two schemes cover all six, worker- and server-side LMO (Algorithms A8–A9,

Optimizer	η_0	**Up**	**Down**	**Rds. to 80%**	**Test acc (%)**
Muon	0.1	32	32	260	85.99 ± 0.26
SignMuon	0.05	**1.09**	**1.09**	300	85.81 ± 0.25
EF21-SignMuon	0.05	**1.09**	32	600	85.16 ± 0.15
MuonUSign	0.05	**1.09**	32	540	84.62 ± 0.17
EF21-MuonSign	0.05	**1.09**	**1.09**	980	83.99 ± 0.28
EF21-MuonUSign	0.05	**1.09**	32	1000	83.75 ± 0.21
SignSGD	0.02	**1.09**	**1.09**	1080	81.83 ± 0.43
MuonSign	0.05	**1.09**	**1.09**	$1300^\dagger$	80.68 ± 1.88
SGD	0.1	32	32	$1367^\dagger$	79.79 ± 1.55
Adam	0.001	32	32	—	77.12 ± 1.04

Table 2: CIFAR-10 federated learning on CNN2, ordered by accuracy: $N = 11$ clients, homogeneous split, 2000 rounds at batch 192, momentum 0.9. Test accuracy is the mean of the final five evaluations over five seeds, $\pm$ one s.d. *across* seeds; Up and Down are bits per parameter per round each way (Appendix A.14). † mean over the seeds reaching 80% in budget (4/5 for MuonSign, 3/5 for SGD); Adam reached it on none.

Table A9).

Federation separates the methods far more, and Table 2 admits three readings. First, SignMuon attains Muon's accuracy at one bit per parameter in *both* directions, within the seed spread, and beats SignSGD at that same budget by several accuracy points, considerably more than centrally. Second, five seeds resolve the three placements, which single-seed evidence had shown as coincident: they order after, before, both sides, the last below SignSGD. Where only the uplink is constrained, the sign-before methods are at their best, MuonUSign and EF21-MuonUSign spending their bit there and broadcasting exactly; EF21-MuonUSign is then the cheapest guarantee available, the rank penalty of Corollary 1

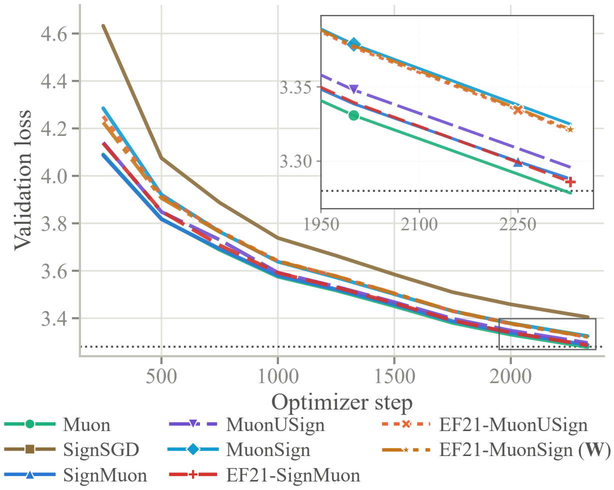

Figure 3: NanoGPT speedrun, 8×H100: validation loss against optimizer step, one run per method; dotted is the target 3.28, and EF21-MuonSign is drawn at its broadcast model $\mathbf{W}$. The inset magnifies the boxed tail, where SignSGD lies above the range shown.

being charged only by a compressed downlink.

Third, and least comfortably, the best placement is the one the theory condemns. Sign-after is what Theorem 1 makes diverge and error feedback repairs for no (L, η, μ) (Theorem 4), yet the sign-after methods are the strongest compressed ones here, centrally, and on nanoGPT, while the two carrying an unconditional guarantee trail SignMuon by several seed spreads. The second compression is not the source of that gap: EF21-MuonSign stands marginally *ahead* of the uplink-only EF21-MuonUSign, so the downlink loop is free here and the placement alone separates the EF21 family from SignMuon. EF21-MuonSign is scored at its server model $\mathbf{X}_t$, not the model its clients hold (13); the two track each other here but not at transformer width (Remark 3, Appendix A.10).

5.3 Language Modelling

We test where the matrices are large enough for the layer-rank dependence of Corollary 1 to take effect: the modded-nanoGPT speedrun (record #40), a 12-layer transformer with 768×3072 hidden matrices, trained on 611M FineWeb tokens on 8×H100 (Jordan et al. 2024a). Each method replaces the record's Muon on the hidden matrices and gates, leaving all else untouched; our Muon run is record #40's own update rule, re-implemented without its Triton kernels, and reproduces its published result to 0.6 of the upstream seed spread. Rates are fixed a priori by the unit-gain rule at one η_0 per *family*, so every contrast is matched-hyperparameter (Appendix A.16).

Three conclusions follow (Table 3, Figure 3). First, the matrix-aware 1-bit methods transfer to language modelling: every one improves on SignSGD by at least 0.08 in validation loss, while the best two, EF21-SignMuon and Sign-

Method	Step	η_0	Val. loss	Steps to 3.35
Muon (record #40)	LMO	0.06	**3.2785**	1.00×
EF21-SignMuon	LMO	0.06	3.2860	1.02×
SignMuon	SIGN	0.03	3.2881	1.02×
MuonUSign	LMO	0.06	3.2959	1.05×
EF21-MuonUSign	LMO	0.06	3.3203	1.13×
EF21-MuonSign ($\mathbf{W}$)	LMO	0.06	3.3213	1.14×
MuonSign	SIGN	0.03	3.3249	1.14×
SignSGD	SIGN	0.03	3.4049	–
EF21-MuonSign ($\mathbf{X}$)	LMO	0.06	5.5198	–

Table 3: NanoGPT after 2330 steps (611M tokens), one run per method. Record #40 reports 3.2780 ± 0.0009 over five seeds, so differences below $\sim$0.003 are noise; "Steps to 3.35" is relative to Muon. The last row is the server model $\mathbf{X}$ of the same run as the $\mathbf{W}$ row.

Muon, trail full-precision Muon by under 0.01 at equal wall-clock. Second, error feedback is inexpensive but not free: MuonUSign → EF21-MuonUSign takes $\sim 8\%$ more steps, the price of steering the LMO with an estimator that lags its target. EF21-SignMuon, whose target $\operatorname{polar}(\tilde{\mathbf{M}}_t)$ is orthogonal and hence isotropic, forfeits almost nothing and is the strongest compressed method here, the condemned placement again finishing ahead. Third, EF21-MuonSign's two models (13) separate at this width: its broadcast model $\mathbf{W}$, at which every gradient is evaluated, is indistinguishable from EF21-MuonUSign, so the compressed downlink is free where training occurs, while the server model $\mathbf{X}$, the iterate the guarantee bounds, settles well above it, a persistent offset that Appendix A.16 localizes to one layer type, the zero-initialized `c_proj`. This is the mechanism of Remark 3 (Appendix A.10), not a tuning failure: only a spectrally contractive downlink compressor removes it.

6 Conclusion

A sign step and a spectral step are each an LMO, sound alone; composed, they are an LMO for no norm, and the descent property is lost in the composition, not in either factor. On minimal explicit instances *every* placement of the sign around the oracle turns the update into an ascent direction on a linear objective, at any step size, and error feedback on the oracle's *output* repairs none of them, its one shared magnitude being unable to track a target that may jump by a constant. Applied to the gradient it does succeed, at a one-bit uplink and, with a second loop on the model, in both directions; the second bit is nearly free in practice but costs a factor in the layer rank in theory. Experiment orders the methods the other way round, which we document rather than resolve: what we preclude is a guarantee holding *everywhere*, not one under a realistic condition. Identifying it is the open question, as are the (L^0, L^1) rate under a compressed downlink and whether *any* one-bit compressor contracts in a layer norm.

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